Who Audits the Audit?

Fairness Audit Infrastructure Encodes Normative Commitments, and They Surface as Penalties on the Right Answer

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ABSTRACT

Fairness audits are becoming gates. They appear in procurement requirements, regulatory bias-audit mandates, and internal model sign-off, and increasingly a system ships or does not ship on the basis of what an audit reports. An audit, however, is software, and software has defaults. We ask what those defaults decide on our behalf.

We built a fairness audit over three lending decisions — consumer credit, mortgage origination, and peer-to-peer default risk — in which a deterministic pipeline computes every metric and assigns every severity label, and a language model supplies only interpretation. Auditing the full cleaned population rather than a held-out split, we find that ordinary, defensible components of the audit stack change its conclusions. Auditing a 2,196-row sample, the pipeline ranked American Indian men as the *best*-treated group in the mortgage data — disparate impact 1.1504, above parity — on the evidence of four applicants who happened all to be approved. Over all 10,978 applicants the same model puts that group at **0.6734**, the worst in the dataset. Small samples do not merely hide disparities; they invert them, and a reporting floor cannot tell the two cases apart. Auditing age at the 40+ threshold borrowed from employment law returns near parity (0.9791) where the threshold credit law actually specifies, 62+, returns the largest age disparity in the data (0.8897). A favorable-label convention inverted a headline ratio from 0.9931 to 0.0. And two detectors we wrote to hold the model to account penalised correct behaviour: one scored terse-but-correct answers as refusals and then zeroed a second provider's evaluation, on the strength of which that provider was permanently blocked in our configuration; the other flagged a newer model five times as often as its predecessor, though 67% of those flags are values derivable from the context the model was given.

None of these components is a bug. Each is a reasonable engineering decision that silently resolves a normative question — which groups are large enough to count, which statute defines a protected class, which outcome is favorable, what counts as an answer. We argue that this is the general condition of automated evaluation rather than a property of our pipeline, and we set out what audit tooling should expose to make these commitments contestable.

CCS CONCEPTS

• **Social and professional topics** → **Governmental regulations**; • **Computing methodologies** → *Machine learning*; • **Human-centered computing** → *Human computer interaction (HCI).*

KEYWORDS

algorithmic auditing, fairness measurement, disparate impact, intersectionality, lending, ECOA, large language models

1 INTRODUCTION

Two things are happening at once. Fairness audits are being written into consequential processes — vendor procurement, regulatory bias-audit requirements, internal model risk sign-off — so that an audit result increasingly determines whether a system is deployed. And the audits themselves are being automated, with metric libraries, severity thresholds, quality checklists and, lately, language models standing in for the judgement a human auditor once supplied.

The combination deserves attention. When an audit was a report written by a person, its assumptions were arguable in the ordinary way. When an audit is a pipeline, its assumptions are defaults: chosen once, applied everywhere, and visible only to whoever reads the source. Lee et al. [22] make the organising commitment of human-centered fair AI that fairness is defined relative to a human decision context rather than in metric space. We take that seriously as a claim about *apparatus*: if a fairness claim depends on its decision context, so does every component that produces it.

This is not speculative. Deng et al. [10] find practitioners using fairness toolkits in ways their designers did not anticipate; Lee and Singh [23] map systematic gaps in what open-source toolkits support; Holstein et al. [17] report that practitioner needs diverge sharply from what the literature supplies. Mei et al. [28] show, in a speech setting, that audit *pipelines* introduce pitfalls that change conclusions, and Zaccour et al. [39] show that quantitative audits are more often blocked by data access than by method. The common thread is that the instrument is part of the finding.

What we did. We built a complete fairness audit over three lending decisions and then examined the audit rather than only its output. The deterministic layer computes every metric with AIF360 [3] and assigns every severity label by fixed rule; a language model supplies interpretation and is scored against that fixed ground truth, so no model grades another model's opinion. Audits run over the full cleaned population — 1,000 and 10,978 rows — with every row scored out-of-fold by a model that never saw it, rather than over a held-out split.

What we found. Six components of an ordinary audit stack changed its conclusions (Table 1). Each is defensible in isolation. Each resolves a normative question without recording that it did so.

Table 1: Results and the evidence for each. Every artifact path is in the released repository; every figure is regenerated by the pipeline rather than transcribed.

Finding	Evidence	Proof
Small samples inverted a disparity	On a held-out split, `American Indian × Male` reads 1.1504 — above parity, the best cell — on four applicants. At full population the same group is **0.6734**, the worst cell in the data. Small samples invert disparities, not merely hide them.	`scripts/compare_populations.py`; `population_comparison.json`
The age audit used the wrong statute	40+ is the ADEA (employment). ECOA protects 62+ (Reg. B §1002.2(o)). Pooling young against senior applicants cancels them: banded `age_group` reads 0.9791 (LOW); the statutory cut reads **0.8897**, the largest age disparity in the data.	`hmda/fairness/fairness_summary.json`; `clean_hmda.py`
A label convention inverted a headline metric	Selection rates computed on the unfavorable label of a target named `loan_default` gave DI = 0.0 (CRITICAL) where the correct value is 0.9931.	`tests/test_lending_club_fairness.py`
A refusal detector measured brevity, then set policy	6 flagged responses, 0 genuine refusals: a 22-character correct answer fell under a 40-character threshold, under a prompt demanding terseness. Applied to a second provider it zeroed 14 of 24 cycles, moving its mean from 70.0 to 19.79; that provider is permanently blocked in our config on that evidence.	`section_f/gemini_hallucination_summary.json`; `research_guardrails.json`
A fabrication detector penalised arithmetic	`gpt-5-mini` tripped 5× more flags than `gpt-4o` on identical input, but 14 of 21 (67%) flagged values are derivable from the provided context (88.88 is 0.8888 as a percentage).	`llm_benchmark/gpt-5-mini/`; `RESULTS.md` §4.1
Prompt sensitivity is model-generation-specific	Cycle-mean spread across four prompt strategies: 14.7 and 9.0 points for `gpt-4o`; 0.0 and 1.9 for `gpt-5-mini`, same packs.	`llm_benchmark_v2/gpt-4o/`; `RESULTS.md` §4

Contributions. (i) An empirical demonstration, on real lending data, that four standard components of an audit stack — a reporting floor, a protected-class threshold, a label convention, and two output detectors — each change what the audit concludes. (ii) A demonstration that small-sample audits invert disparities rather than only hiding them, isolated by holding the model fixed and varying only the audited population. (iii) A cross-model result showing that the prompt-sensitivity finding widely reported for LLM evaluators is specific to a model generation. (iv) A released, cost-capped, offline-reproducible audit pipeline in which every claim above is regenerable.

2 RELATED WORK

Auditing, and auditing the auditors. Metaxa et al. [29] survey algorithm auditing as method; Raji and Buolamwini [32] show external audits change vendor behaviour, and Raji et al. [33] give an internal end-to-end framework. Kroll [21] argues traceability is what makes accountability operational — the stance our frozen-artifact design takes. Documentation interventions [15, 30] and process interventions [27] share our premise that fairness work needs infrastructure, and share the exposure we report: a checklist is a mechanical proxy for judgement, and §4.4 is a case study in what that costs. Mei et al. [28] is the nearest prior work; Zamfirescu-Pereira et al. [40] diagnose human error inside an image-analysis pipeline. We add a lending case where the pitfalls are a size threshold and a statutory misattribution, and where the affected quantity is which group the audit names as most harmed.

Measurement. That fairness criteria conflict is settled [7, 20], and that choosing among them is not a technical act is argued by Selbst et al. [35], whose abstraction traps describe our setting. Corbett-Davies and Goel [9] catalogue how fairness measures mislead detached from decision context. Barocas and Selbst [1] and Wachter et al. [37] establish that legal discrimination does not reduce to a metric threshold, which is why we treat the four-fifths screen as a screening signal and make no legal claim. On intersectionality, Buolamwini and Gebru [5] give the canonical empirical demonstration that marginal accuracy hides joint-group failure and Foulds et al. [14] the formal definition; §4.1 contributes the observation that a routine size floor removes exactly those joint groups from the report.

Whose criteria. A growing line asks who should choose fairness criteria rather than how to compute them: Luo et al. [25] develop a protocol for negotiating metrics with stakeholders, Luo et al. [26] document how much stakeholders disagree on identical evidence, Luo et al. [24] extend this to affected individuals, and Deshpande and Sharp [11] ask who the stakeholders are at all. Stumpf et al. [36] argue fairness assessment needs user-centred evaluation; Kim et al. [19] find the construct of equity fragmenting under practical demand. Our contribution is adjacent: this literature asks who sets the criteria, and we show that the apparatus resolves several such questions before anyone is asked.

Language models as evaluators. Using an LLM to interpret metrics invites the LLM-as-judge critique [41]. Our design avoids the worst of it: ground truth is a deterministic rule over computed metrics, never another model's opinion. Wester et al. [38] study when and how LLMs deny user requests, which is precisely the phenomenon our refusal detector was built to catch and, as §4.4 shows, failed to measure.

3 WHAT WE BUILT

3.1 Three lending decisions

The three settings differ along the two axes that most change what an audit can conclude: whether the protected attribute is a protected class or an economic characteristic, and whether the classifier discriminates enough for disparity to be measurable. We use German Credit [16], HMDA Georgia [8], and a Lending Club default task. Lending discrimination is documented empirically across this span — appearance and gender effects in peer-to-peer markets [12, 31], FinTech-era consumer lending [2], local credit-market conditions [6], and historical records used as training data [34].

3.2 Deterministic ground truth

For each protected attribute we compute disparate impact, demographic parity difference, equal opportunity difference, average odds difference and the Theil index with AIF360 [3], cross-checked against Fairlearn [4]. A fixed rule maps disparate impact to a severity band, with the CRITICAL boundary at 0.72 and the four-fifths screen at 0.80. The model never recomputes a number; asserting an unprovided value is recorded as fabrication, not as a result.

3.3 Auditing the population, not a sample

Audits here run over every cleaned row, with predictions generated by stratified five-fold cross-validation so each applicant is scored by a model that never saw them. This matters more than it sounds: at held-out-split size the HMDA `race×sex` cells for American Indian, Multiracial and Pacific Islander applicants hold three to six people each. Those groups do not become measurable through a better estimator; they become measurable through more rows.

3.4 Disclosure

None of the three models would pass a real fair lending review, for reasons orthogonal to our claims, and we state this rather than let it be discovered: the German Credit model takes applicant sex and age as input features; the Lending Club `gender` attribute is synthetic, assigned by an independent coin flip, so that setting contains no protected class; and HMDA's public data carry no credit score, the primary underwriting variable. We audit these systems as instruments for studying audit infrastructure, not as candidate lending models.

4 RESULTS

We report marginal disparate impact for every protected attribute in Table 2 as the baseline against which the rest of this section should be read. On these numbers alone, the audit is unremarkable: every attribute in every setting clears or nearly clears the four-fifths screen, and nothing demands attention. Each subsection below concerns a component of the audit that determined one of these numbers, or determined which numbers appeared at all.

Table 2: Marginal disparate impact, full population. Read alone, this table reports an unremarkable audit. †The ECOA cut; `age_group` is the banded view of the same applicants. ‡Synthetic attribute (§3).

Setting	Attribute	DI
German Credit	`Sex`	0.8888
	`AgeGroup`	0.9258
	`foreign_worker`	0.8387
HMDA (Georgia)	`race`	0.9371
	`sex`	0.9572
	`age_group` (banded)	0.9791
	`age_62_plus`†	**0.8897**
Lending Club	`gender`‡	0.9931
	`income_level`	1.0072
	`loan_amount_level`	1.0063

Table 3: HMDA `race` × `sex` under an $n \geq 15$ floor. Identical model; only the audited population differs.

	Held-out split (n=2,196)	Full population (n=10,978)
Cells suppressed	**5 of 11**	1 of 12
Worst cell *reported*	0.8169 (MOD.)	**0.6734** (CRIT.)
`Am. Indian×Male`	**1.1504** (best cell)	**0.6734** (worst cell)
`Am. Indian×Female`	0.1917 (n=6, noise)	0.8815 (n=25)

4.1 Small samples invert disparities, and the floor cannot help

Subgroup reporting floors are standard and defensible: a ratio computed on six people is not evidence. Ours suppresses cells below $n = 15$. To isolate what that costs, we hold the model fixed and vary only the audited population (Table 3).

The failure is sharper than suppression. On the split, `American Indian × Male` appears at DI = 1.1504 — *above* parity, the best-treated cell in the table. Four applicants, four approvals. At full population that same group is at 0.6734 with n=31: the worst cell in the dataset, in the CRITICAL band. The audit did not miss the disparity; it **inverted** it, and attached a reassuring number to the inversion.

The floor was right to suppress what it suppressed. `American Indian × Female` reads 0.1917 on the split — one approval in six — which looks like the most severe disparity in the study and is in fact noise: at full population the same cell is 0.8815 (n=25). A reader who overrode the floor to surface that number would have reported a catastrophe that does not exist.

That is the uncomfortable shape of this result. The floor is not the error and removing it is not the fix. At this sample size the estimates for small groups are unreliable *in both directions* — one group looked catastrophic and another looked flawless, and both were wrong. Every affected cell belonged to a racial minority, which is mechanism rather than coincidence: a floor on group size is a floor on group population, so the groups whose treatment is least certain are exactly the groups least likely to be counted.

Table 4: HMDA `race` × `sex`, full population. The floor suppresses the starred row. On a held-out split it suppressed the top five.

race	sex	n	sel. rate	DI
American Indian	Male	31	0.5806	**0.6734**
Multiracial	Female	18	0.6111	0.7088
Multiracial	Male	29	0.6552	0.7599
Pacific Islander	Female	12*	0.6667	0.7732
Pacific Islander	Male	15	0.7333	0.8505
Black	Female	1,994	0.7442	0.8632
Black	Male	1,707	0.7452	0.8643
American Indian	Female	25	0.7600	0.8815
White	Female	2,318	0.7865	0.9121
White	Male	4,019	0.8281	0.9604
Asian	Male	556	0.8525	0.9888
Asian	Female	254	0.8622	1.0000

At full population the same floor suppresses one cell of twelve and the worst disparity, `American Indian × Male` at 0.6734 (n=31), becomes visible. Table 4 gives the complete cell table, which we print in full because the argument of this paper is precisely that partial tables mislead. The three worst cells hold 31, 18 and 29 applicants and would be invisible under the split-sized audit; the two cells an audit typically discusses, `Black × Female` and `Black × Male`, sit sixth and seventh.

The floor is not a bug and we do not propose removing it. The point is narrower and harder to design around: a threshold chosen for statistical prudence distributes its entire cost onto the smallest groups, which in lending are generally the groups least able to absorb it, and it does so silently — the audit output lists what passed, not what was withheld. Our own codebase carried two different floors, $n \geq 15$ and $n \geq 30$, in adjacent modules, so the same cell was reportable or not depending on which one read it.

4.2 The age audit used the wrong statute

Fairness work on credit routinely audits age at 40+, and ours did. That threshold is from the Age Discrimination in Employment Act, which governs employment. Credit is governed by the Equal Credit Opportunity Act, whose Regulation B §1002.2(o) defines the protected class as **62 or older**. A comment in our own cleaning code asserted that 40+ "aligns with US ECOA". It does not.

The substitution is not cosmetic. Our banded age attribute pools young and senior applicants against the middle band; the two move in opposite directions, so pooling cancels them. The banded attribute returns DI = 0.9791, severity LOW — a clean bill of health. Audited on the statutory cut, the same model over the same rows returns **0.8897**, the largest age disparity in the dataset and the lowest of its four protected attributes. The flag required, `applicant_age_above_62`, was present and unused in the raw HMDA file; it is published precisely because §1002.2(o) turns on it, and it is necessary because the public age bands straddle 62, so no banded attribute can isolate the protected class.

One asymmetry this exposes and cannot resolve: §1002.6(b)(2) expressly permits *favouring* an elderly applicant. Disparate impact is symmetric, so a ratio above 1 here is lawful and is not a reverse-disparity finding, yet the metric reports it identically. The instrument cannot express the rule it is being used to check.

4.3 A label convention inverted a headline metric

In the Lending Club setting, `gender` initially returned DI = 0.0 and a CRITICAL label — the most severe output the pipeline can produce. It was an artifact. The target is `loan_default`, so label 1 is the *unfavorable* outcome; selection rates computed on predicted default give 0.0000 and 0.0069, hence DI = 0.0. Oriented on the favorable outcome the rates are 1.0000 and 0.9931, so DI = 0.9931: near parity.

Nothing was misused. AIF360 computed what it was asked. The data were clean. Disparate impact was the right screen. The error lives entirely in the mapping from a column name to a normative category, and that mapping is a convention rather than a computation: a target named `approved` makes label 1 favorable, a target named `loan_default` inverts it, and nothing in the metric signature records which case holds. Default, delinquency, charge-off and fraud are all conventionally coded with the adverse event as 1, so fairness tooling applied to any risk model inherits this unless an analyst supplies the mapping. It is now a required argument and a regression test in our pipeline, which converts a silent inversion into a build failure.

4.4 Our own detectors penalised correct behaviour

The findings above concern conventions we inherited. These concern code we wrote, deliberately, to hold the model to account.

The refusal detector measured brevity. It flagged six of nine responses under a terse-output prompt strategy. None was a refusal: no flagged record contained a refusal phrase, a missing field, or an invalid severity. The detector treats any narrative field under 40 characters as effectively empty, so `how_to_fix: "DisparateImpactRemover"` — correct, on-spec, and naming the applicable post-processor [13] in 22 characters — registered as a refusal. Compliance with the instruction was the trigger. Reporting the raw output would have yielded a 66.7% refusal rate and a tidy story about constrained prompting inducing evasion, entirely an artifact of our instrument.

And then it set policy. Applied to a second provider, the same detector flagged 19 of 24 cycles and mechanically zeroed 14 of them. Every zero-score cycle is a flagged cycle; excluding them moves that provider's mean score from 19.79 to **70.0**. At least five flagged cycles are not refusals — one scored 85/100. On the strength of the 19.79 figure, that provider was recorded as permanently blocked in our guardrail configuration, where the block persists. An invalid measurement did not merely mislead a table; it produced a durable governance decision about a vendor.

The fabrication detector penalised arithmetic. Our second detector flags any numeric value absent from the provided context. Replaying identical packs against two model generations, `gpt-5-mini` trips 17 flags across 28 calls where `gpt-4o` trips 4. Read naively, the newer model fabricates five times as often. Checking each flagged value against the context, **14 of 21 (67%) are derivable from it**: 88.88 is the disparate impact 0.8888 restated as a percentage, 83.87 is 0.8387, and most of the remainder are gaps between two provided values expressed in percentage points. The detector matches raw float literals, so converting a ratio to a percentage or subtracting two given numbers registers as fabrication. The model that reasons more explicitly is ranked as the less trustworthy one, precisely because it shows its work.

A guardrail punished "collect more data". A third check requires every proposed mitigation to name a canonical algorithm — reweighing [18], disparate impact removal [13], and similar. It failed exactly the two attributes whose problem *is* data scarcity, where the audit proposed acquiring records to a stated floor, hierarchical shrinkage and manual review. No post-processor repairs a six-record subgroup. The gate encodes an assumption that every fairness problem has an algorithmic mitigation, and where that is false it penalises the only defensible answer.

We left all three unrepaired and report them instead. Repairing them would have concealed the finding, and the finding is the point: three instruments, written in good faith by people who understood the domain, each scoring a correct behaviour as a fault.

4.5 Prompt sensitivity is model-generation-specific

A widely reported result about LLM evaluators is that prompt phrasing moves the verdict. It reproduces here, and it does not generalise across model generations. On identical packs, the spread in cycle-mean score across four prompt strategies is 14.7 and 9.0 points for `gpt-4o` and **0.0** and **1.9** for `gpt-5-mini`. Claims of the form "LLM auditors are unstable under paraphrase" should therefore carry a model and a date.

Two methodological notes. `gpt-5-mini` rejects the `temperature` parameter outright, so a temperature sweep is impossible on it and any repeated-sampling study must use another model — an API constraint that silently determines which experiments a benchmark can run. And our four-cycle design varies prompt strategy rather than resampling one strategy, so cycle-to-cycle differences confound phrasing with sampling noise; the cross-model contrast above is robust to this, since both models face the same confound.

5 WHY THIS MATTERS

5.1 Audits are becoming gates, and nobody audits the auditor

The findings above would be curiosities if audits were advisory. They are not. Bias-audit requirements are entering regulation, procurement, and internal model governance, which means an audit result increasingly decides deployment. Our audit, run competently on clean data with a standard toolkit, reported the wrong most-harmed group on a mortgage dataset and issued a clean bill of health on the one age class US credit law protects. Both outputs would have passed review, because both are what the pipeline says.

The asymmetry that matters is directional. A false alarm costs reviewer time. A false clearance closes an open question, and §4.1 and §4.2 are both false clearances — exactly the direction least likely to prompt a second look.

5.2 A single benchmark number can be an artifact of task difficulty

Our headline agreement figure between model and ground truth was 75%. It decomposes to 33.3%, 91.7% and 100%, and the 100% comes from the setting where the audited classifier barely discriminates at all, so agreeing costs nothing. The ordering tracks how hard each case is, not how capable the model is. This generalises past fairness: any benchmark averaging over heterogeneous instances can report a number dominated by the difficulty mix of its constituents rather than by the capability it claims to measure.

5.3 Every mechanical proxy for judgement is a policy

A length threshold that treats brevity as refusal is a policy about what counts as an answer. A keyword list that treats "acquire more data" as a non-mitigation is a policy about what counts as a remedy. A float-literal match that treats a percentage as a fabrication is a policy about what counts as evidence. None was labelled a policy; each was a heuristic written to make an evaluation automatable, and each punished the better answer in the cases where judgement was most needed. As evaluation automates further — LLM judges, automated red-teams, compliance scoring — this class of failure scales with it.

5.4 A workable division of labour

The architecture that survived our evidence is narrow and, we think, transferable: deterministic code owns the numbers and the thresholds; the model owns the interpretation; and the interpretation is scored against the numbers rather than against another model. Calculator, not oracle. The pattern extends to any setting where a model is asked to interpret computed quantities under consequence, and it is the reason we can state which of our own results are artifacts — the ground truth did not move when the model did.

6 WHAT TO DO ABOUT IT

These follow directly from the results and are cheap to adopt.

(1) **Make the favorable outcome an explicit, required argument.** Never infer it from label values. One parameter converts a class of silent inversions into a build failure (§4.3).
(2) **Report what the floor withheld.** An audit should print suppressed cells with their n and their unreported ratio, not omit them. The audit in §4.1 was not wrong about any number it computed; it was wrong about what it did not print.
(3) **Bind protected-class definitions to the governing statute** in code, with the citation next to the threshold. A comment asserting the wrong statute survived review precisely because it looked authoritative (§4.2).
(4) **Treat every mechanical gate as a hypothesis to falsify**, starting with inputs where the correct answer is unusual: the terse answer, the tiny subgroup, the model that shows its work.

(5) **Audit the population, not a convenience sample.** Most of our small-cell instability came from auditing a held-out split rather than every applicant in the review period.
(6) **Record which reference group a ratio uses.** Comparing against the most-favoured cell rather than a designated control group moved one verdict across the four-fifths line, and nothing in the metric signature recorded which convention was in force.

7 WHAT THIS DOES NOT SHOW

Scale. Three datasets, two primary models. These are mechanism-level findings: we show that a class of failure occurs and how it operates, not how often it occurs in the wild. The mechanisms transfer even where the statistics do not — a convention default that inverts a metric is a class of defect, not an incident.

No human subjects. We make claims about what an audit reader would conclude without testing them on audit readers. That is the natural next study and the main gap between this work and the stakeholder line it builds on [24, 25].

No causal or legal claims. All metrics are associational; root-cause mapping is rule-based inference over metric values. ECOA, Regulation B and the four-fifths rule appear as decision context. A disparate impact ratio below 0.80 is a screening signal, not a finding of unlawful discrimination [1, 37].

Model limitations. The German Credit classifier uses protected attributes as features; the Lending Club protected attribute is synthetic; HMDA lacks credit score. Each is disclosed in §3 and each bounds what the corresponding audit can mean.

Sampling confound. Temperature and seed are unpinned, so within-model cycle differences mix prompt strategy with sampling noise.

8 CONCLUSION

We set out to measure whether a language model can audit fairness. The more durable result concerns the measuring apparatus. A sample too small decided that the worst-treated racial group in a mortgage dataset was the best-treated one. A threshold borrowed from employment law decided that the age class credit law protects was fine. A column name decided whether a disparity ratio read 0.9931 or 0.0. A character count decided that a correct answer was a refusal, and then decided that a vendor should be blocked.

None of these was a bug, and none was the work of anyone careless. They are what it looks like when normative questions are answered by defaults. As fairness audits become gates, the defaults become policy — and unlike policy, they are not written down anywhere a reviewer would think to look. The remedy we can offer is unglamorous: make the commitments explicit, print what was withheld, and test the instruments on the cases where the right answer is unusual, because that is where they fail.

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